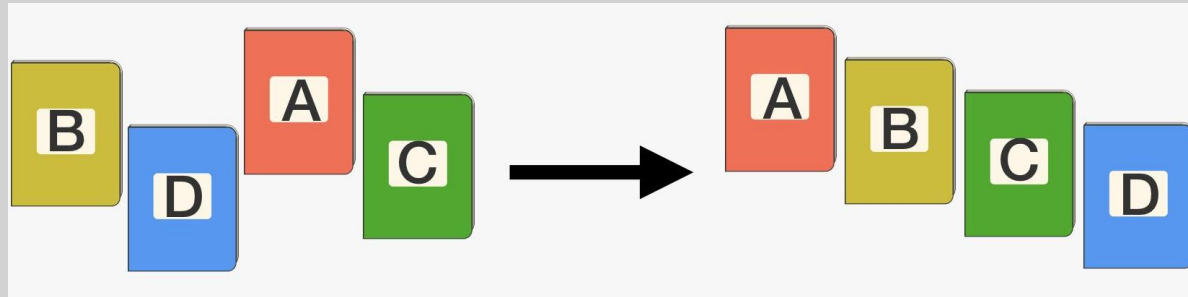


Selection Sort

MODULE 3

Introduction to Sorting

- Sorting is arranging the elements in ascending and descending order



Procedure to perform selection sort

- Let us an Array A of N elements is in memory.
- Find the smallest or biggest element from a given list of element and place it in the first position and find the second smallest or biggest element in the remaining list and place in the second position and so on...

This is the algorithm, now you carry on

- Now you take pen and paper. Do the following sorting (Selection sort) using algorithm then show me through webcam the steps you have written on the paper.
- Example list
- 45, 78, 12, 54, 15, 23, 56

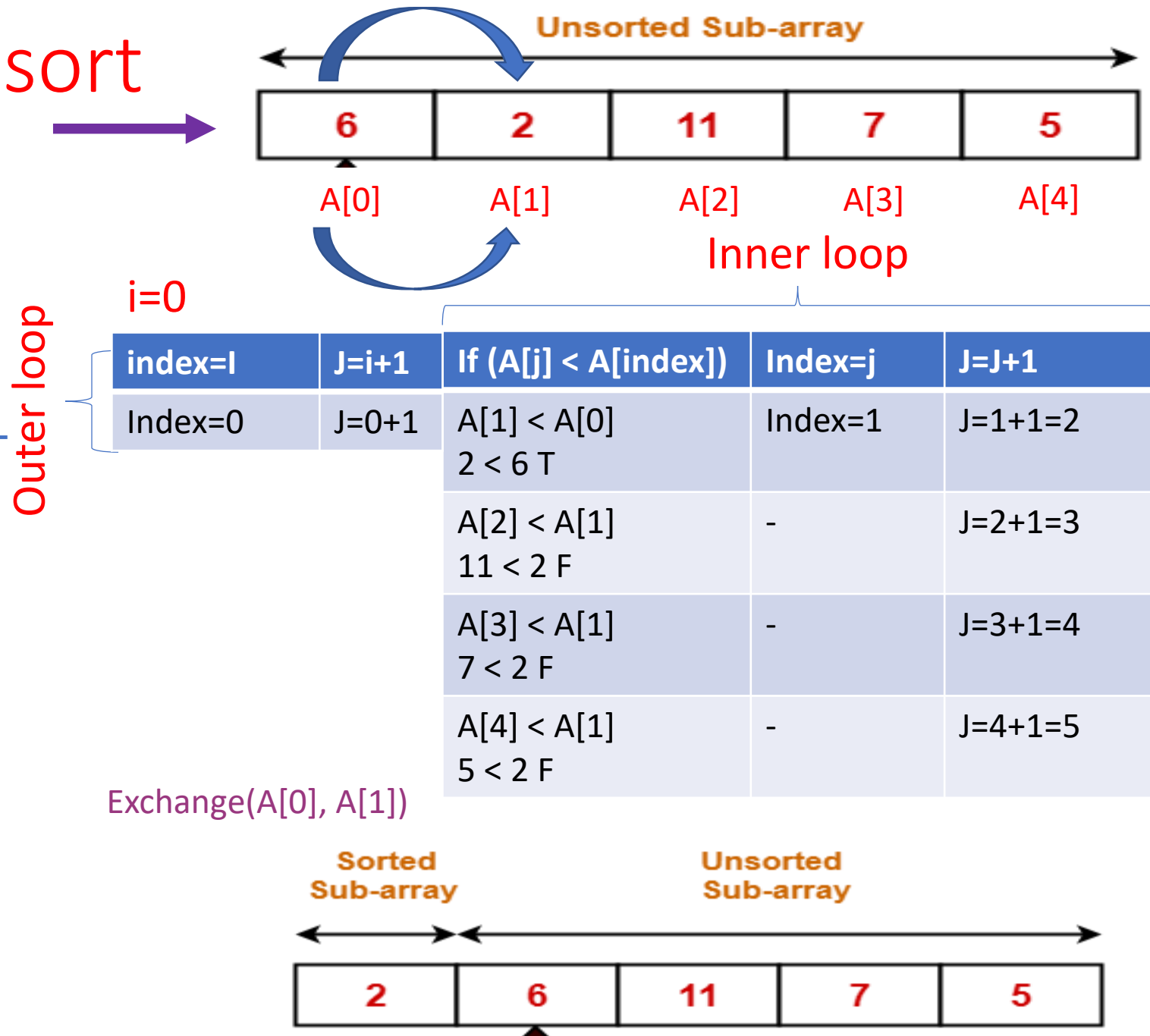
- Selection(A,MAX)
- Repeat for i: = 1 to MAX
- Do
- Set Index=i
- Repeat for j := i+1 to MAX
- Do
 - If $A[j] < A[index]$
 - then
 - set index = j;
 - End If.
- Done
- Exchange($A[i]$, $A[index]$)
- Done

Algorithm Selection sort

- Selection(A,MAX)
- Repeat for i: = 1 to MAX
- Do
- Set Index=i
- Repeat for j := i+1 to MAX
- Do
- **If** A[j] < A[index]
- **then**
- set index = j;
- **End If.**
- Done
- Exchange(A[i], A[index])
- Done

Algorithm Selection sort

- Algorithm Selection(A,MAX)
- Repeat for i: = 0 to MAX-1
- Do
 - Set Index=i
 - Repeat for j := i+1 to MAX-1
 - Do
 - If $A[j] < A[\text{index}]$
 - then
 - set index = j;
 - End If.
 - Done
- Exchange($A[i]$, $A[\text{index}]$)
- Done



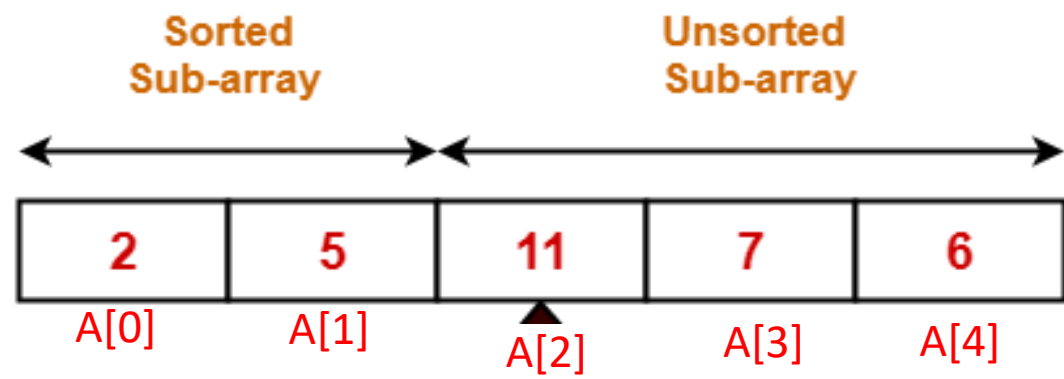
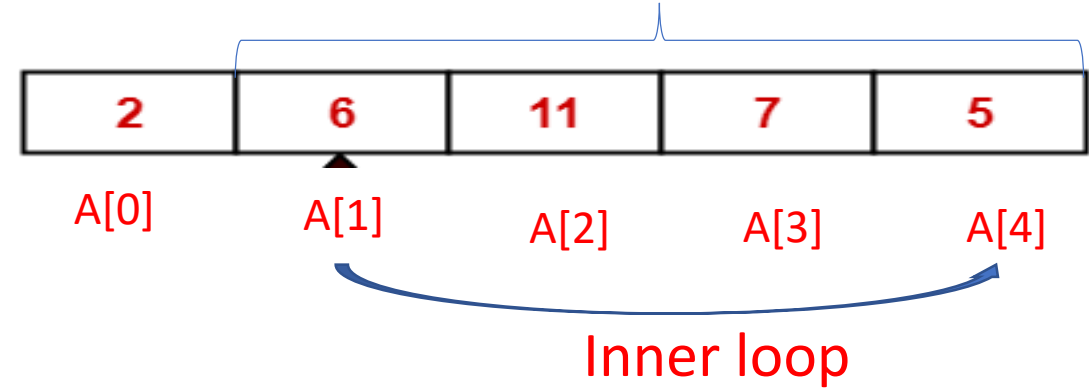
Algorithm Selection sort

- Algorithm Selection(A,MAX)
- Repeat for i: = 0 to MAX-1
- Do
 - Set Index=i
 - Repeat for j := i+1 to MAX-1
 - Do
 - If $A[j] < A[index]$
 - then
 - set index = j;
 - End If.
 - Done
- Exchange($A[i]$, $A[index]$)
- Done

Outer loop

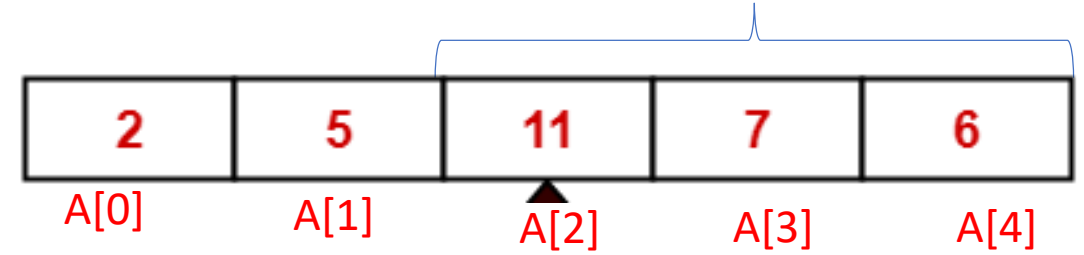
i=1				
index=i	J=i+1	If ($A[j] < A[index]$)	Index=j	J=J+1
Index=1	J=1+1	$A[2] < A[1]$ 11 < 6 F	-	J=2+1=3
		$A[3] < A[1]$ 7 < 6 F	-	J=3+1=4
		$A[4] < A[1]$ 5 < 6 T	Index=4	J=4+1=5

Exchange($A[1]$, $A[4]$)



Algorithm Selection sort

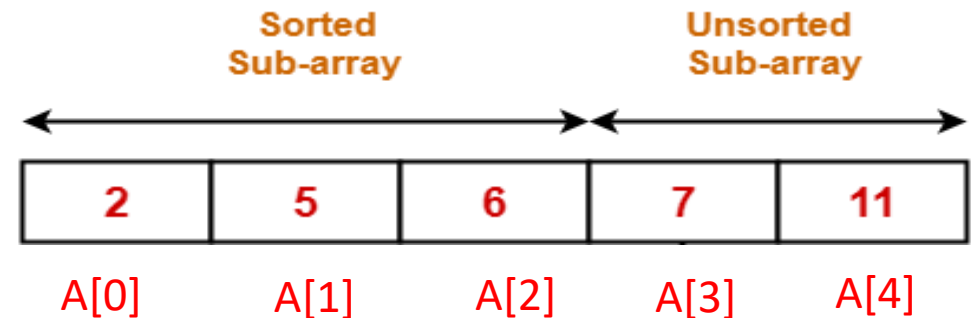
- Algorithm Selection(A,MAX)
- Repeat for $i := 0$ to $MAX-1$
- Do
 - Set $Index=i$
 - Repeat for $j := i+1$ to $MAX-1$
 - Do
 - If $A[j] < A[index]$
 - then
 - set $index = j$;
 - End If.
 - Done
- Exchange($A[i]$, $A[index]$)
- Done



Inner loop

Outer loop	i=2				
	index=l	J=i+1	If (A[j] < A[index])	Index=j	J=J+1
	Index=2	J=2+1	A[3] < A[2] 7 < 11 T	Index=3	J=3+1=4
			A[4] < A[3] 6 < 7 T	Index=4	J=4+1=5

Exchange($A[2]$, $A[4]$)



Algorithm Selection sort

- Algorithm Selection(A,MAX)
- Repeat for i: = 0 to MAX-1
- Do
 - Set Index=i
 - Repeat for j := i+1 to MAX-1
 - Do
 - If $A[j] < A[\text{index}]$
 - then
 - set index = j;
 - End If.
 - Done
- Exchange($A[i]$, $A[\text{index}]$)
- Done



2	5	6	7	11
A[0]	A[1]	A[2]	A[3]	A[4]

Outer loop

i=3

index=i	J=i+1	If ($A[j] < A[\text{index}]$)	Index=j	J=J+1
Index=3	J=3+1	A[4] < A[3] 11 < 7 F	-	J=4+1=5

Inner loop

Exchange($A[3]$, $A[3]$)

Sorted array

2	5	6	7	11
A[0]	A[1]	A[2]	A[3]	A[4]

Time Complexity

- It has $O(n^2)$ time complexity.



End